

Wei Cheng (Wayne) Chiu

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SUMMARY

Machine learning engineer working the whole depth of the LLM inference stack — hand-written CUDA / Tensor-Core kernels underneath, TensorRT-LLM / vLLM / SGLang serving above, and customer deployments on top. Currently a **Field Application Engineer** at Taiwan AILabs, putting on-premises LLM systems (FedGPT) into customer environments. Previously **sole developer** of an enterprise multi-agent AI platform on DGX-class hardware, taken from PoC to production at **two enterprise customers**. Active upstream — **17 merged/landed patches, 60 in review** across FlashInfer, vLLM, SGLang, PyTorch, Dynamo, CUTLASS, TensorRT-LLM and LMDeploy, concentrated on early consumer-Blackwell (**SM120**) and **NVFP4** enablement. Research background in AI/LLM security and privacy-preserving ML.

TECHNICAL SKILLS

LLM Serving & GPU	vLLM, TensorRT-LLM, SGLang, Triton, NVIDIA NIM, Dynamo, FlashInfer, NCCL, Flash Attention, KV-cache, Quantization (FP8 / INT4 / NVFP4); CUDA C/C++, WMMA & <code>mma.sync</code> Tensor Cores, CUTLASS, Nsight Systems & Compute
LLM Apps & Agents	Multi-Agent Orchestration (LangChain, LangGraph), Hybrid RAG (Vector + Keyword + Knowledge Graph), LLM Evaluation & Guardrails, Function Calling / Tool Use, MCP, Coding Agents
Infrastructure & Ops	Linux, Kubernetes, Docker, On-prem GPU Serving, Distributed Tracing; H100, DGX Spark (GB10), Blackwell workstation hardware
Privacy & Security ML	Federated Learning, Differential Privacy, Secure Multi-Party Computation (MP-SPDZ), Safety Alignment
Languages	Python, TypeScript, Rust, C/C++; PyTorch, FastAPI, NestJS; PostgreSQL, Milvus, Redis

EXPERIENCE

Taiwan AILabs <i>Field Application Engineer</i>	Jul 2026 – Present <i>Taipei, Taiwan</i>
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- Field deployment of on-premises LLM systems (**FedGPT**) into customer environments — Linux / Kubernetes operations, GPU serving, and customer-facing troubleshooting

SYNCROBOTIC <i>Machine Learning Engineer — Multi-Agent AI, RAG, LLM Serving</i>	Sep 2025 – Jun 2026 <i>New Taipei City, Taiwan</i>
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- Sole developer** of an enterprise multi-agent AI platform on DGX-class hardware — owned end-to-end from PoC through production and rolled out at **two enterprise customers**, partnering with each through deployment and troubleshooting
- Scoped ambiguous enterprise requirements into a working multi-agent system and **unified local and commercial LLM backends** under one routing layer, removing per-team model wrappers
- Designed a planning–execution–validation orchestration with topological scheduling and parallel agent dispatch for multi-step workflows
- Built a hybrid retrieval layer on LightrAG (vector + keyword + knowledge graph) with embedding-based semantic caching, cutting redundant LLM calls and keeping answers consistent on repeat queries
- Owned production reliability — prompt/pipeline regression tests, output validation across quality and safety, distributed tracing; added CUDA-accelerated hot paths on critical inference

Advantech <i>AI Agent Engineer (Intern)</i>	Jun 2025 – Aug 2025 <i>Taipei, Taiwan</i>
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- Built coding agents for internal developer tooling using **function calling / tool use** — automating code review, test generation, and static analysis across the SDLC; benchmarked tool-use reliability across commercial and local LLM backends and designed reusable prompt templates adopted by the team

TSMC (Taiwan Semiconductor Manufacturing Co.) <i>R&D Alternative Military Service — Facility Engineer</i>	May 2021 – Oct 2022 <i>Tainan, Taiwan</i>
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- Supported R&D fab facility operations at the world's leading semiconductor foundry under the national R&D alternative military service program

OPEN SOURCE

SM120 / NVFP4 enablement across the LLM-inference stack <i>FlashInfer · vLLM · SGLang · PyTorch · Dynamo · CUTLASS · TensorRT-LLM · LMDeploy</i>	17 merged/landed, 60 in review prs.wayne.is-a.dev
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- Kernel-level support for consumer Blackwell (**SM120**) and **NVFP4** carried upward through kernels → engines → disaggregated serving

- Focus on *silent-correctness* bugs — the ones where tests pass and the answers are still wrong

RESEARCH & PUBLICATIONS

SelGrad: Selective Gradient Projection for Efficient Safety Alignment Against Harmful Fine-Tuning — first author, with Shao-Jui Wang. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, under review, 2026. Preserves LLM safety alignment under harmful/adversarial fine-tuning at lower compute than full re-alignment. Research at NTUST (advised by Prof. Shao-Jui Wang) in **AI/LLM security** and **privacy-enhancing ML** — federated learning, differential privacy, and secure multi-party computation.

EDUCATION

National Taiwan University of Science and Technology (NTUST) <i>M.S. in Computer Science and Information Engineering — GPA 4.09</i>	Aug 2024 – Apr 2026 <i>Taipei, Taiwan</i>
National Taiwan University (NTU) <i>M.S. in Structural Engineering (2020) B.S. in Civil Engineering (2018)</i>	Sep 2014 – Jun 2020 <i>Taipei, Taiwan</i>

CERTIFICATIONS & AWARDS

NVIDIA DLI — Accelerated Computing with CUDA C/C++ and CUDA Python (5 certificates) · NICS Privacy-Enhancing Technologies (PETs) Competition — Rank 3/11 (DP synthetic data via MP-SPDZ) · International ICT Innovative Services Awards, 2024